AUTOMATED ENVIRONMENTAL MONITORING

User Manual

Ceramic Pressure Sensor Type 0420

Submersible probe for water level and gauge measurement

TerraTransfer GmbH
Ottostraße 19a · 44867 Bochum · Germany
Phone: +49 2327 83 44 85-1
www.terratransfer.de

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I Disclaimer

The information contained in this manual reflected the state of the art at the time of publication. Subsequent updates are possible. This manual does not cover all details of the design, production or variants of the device, nor does it cover every possible situation that may arise during installation, operation or maintenance.

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II Safety Instructions

Read this user manual including all operating instructions completely before installing, connecting and commissioning the Ceramic Pressure Sensor Type 0420. This manual describes the intended operation of the product. It is intended for qualified specialist personnel familiar with installation, assembly, wiring, commissioning and operation.

> **Note:** Keep this manual in a safe place for future reference and pass it on to third parties when transferring the device.

Symbols Used

> **Caution:** Indicates possible impairment of the device or measurement that may result from non-compliance.

> **Note:** Provides useful additional information or recommendations.

General Safety Rules

- Inspect the device for visible damage before commissioning. Do not commission damaged devices.
- Use the sensor exclusively within the limit values specified in the technical data.
- Unauthorised modifications to the device are not permitted and will void the warranty.
- Applicable safety regulations must be observed when working on electrical connections.

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III Specific Safety Instructions

Ceramic Membrane

Caution: The ceramic pressure membrane is pressure-resistant and scratch-resistant. Clean the membrane with clean water and a soft brush (e.g. toothbrush). Do not use abrasives, solvents, sharp tools or high-pressure cleaners.

Capillary and Venting Cable

The sensor operates with relative pressure. The venting capillary integrated in the cable provides the atmospheric reference. Ensure that the cable outlet at the logger end of the capillary always remains dry and air-permeable.

Caution: If the capillary is pinched, kinked or filled with water, the sensor will deliver incorrect readings. Always route the cable without tension and with sufficient bending radius.

Electrostatic Discharge (ESD)

The sensor contains electrostatically sensitive components. Observe standard ESD protection measures when connecting (potential equalisation, do not touch connector contacts without earthing).

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Part 1 Scope of Delivery

Standard Delivery

- Ceramic Pressure Sensor Type 0420 (including connection cable in ordered length, with venting capillary)
- This User Manual

> **Note:** The connection cable is permanently fitted to the sensor. For other cable lengths, please contact TerraTransfer.

Optional Accessories

- Sensor cage for protection against mechanical damage and fouling
- Extension cable (max. total length: 500 m on the SDI-12 bus)
- Aquatos Web LTX, Aquatos mini (compatible data loggers)

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Part 2 Introduction

2.1 Field of Application

The Ceramic Pressure Sensor Type 0420 is a submersible probe for the continuous measurement of water levels and gauges. Typical applications include:

- Groundwater monitoring wells and boreholes
- Surface water: rivers, lakes, ponds
- Pressure pipelines and tanks
- Flood early warning systems

The sensor is particularly suited for use in aggressive or mineral-rich waters where piezoresistive stainless steel membranes are subject to increased wear.

2.2 Intended Use

The Ceramic Pressure Sensor Type 0420 is intended exclusively for the use described in the previous section. Any use beyond this is considered non-intended. TerraTransfer GmbH accepts no liability for damage resulting from non-intended use.

Caution: The sensor is not approved for use in potentially explosive atmospheres (ATEX).

2.3 Features and Operating Principle

The Type 0420 measures the hydrostatic pressure of a water column using a ceramic pressure membrane. The measured relative pressure (difference to atmospheric pressure via the capillary in the cable) is converted into a digital SDI-12 data value that is polled by the connected data logger.

In addition, the sensor captures the water temperature and provides both measured values via the SDI-12 interface.

The integrated TerraTransfer sensor platform also enables wireless configuration and diagnostics via Bluetooth 5 (BLE) using the Bluetooth app.

Note: Bluetooth communication is intended exclusively for configuration and diagnostics. Measurement data transmission to the data logger always takes place via SDI-12.

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Part 3 Installation

3.1 Mechanical Design and Dimensions

The sensor is housed in a cylindrical stainless steel enclosure and fits into standard measurement tubes from ¾″ inner diameter. The connection cable is permanently fitted at the upper end of the sensor in the ordered length.

Property	Value
Housing material	Stainless steel
Membrane material	Ceramic
Cable	Ordered length, permanently fitted, with venting capillary
Protection class	IP68

3.2 Installation in the Measurement Tube

1. Check the measurement tube for deposits and obstructions.
2. Feed the cable downward through the measurement tube until the sensor reaches the desired installation depth.
3. Secure the cable at the tube rim using a suitable cable guide or clamp. Ensure sufficient bending radius (min. 5 cm) and tension-free routing.
4. Note the installation depth (distance from sensor membrane to overflow edge) for later configuration.

> **Caution:** The sensor must not rest on the tube bottom or be clamped. Minimum clearance from the tube floor: 10 cm.

3.3 Cable Connection to the Data Logger

The connection cable of the Type 0420 is designed as an SDI-12 three-wire bus (VCC, Signal, GND). The wiring at the cable end:

Wire	Colour / Function
1	White — Supply voltage (VCC, 5–16 V DC)
2	Green — SDI-12 data signal
3	Yellow — GND

Connect the cable according to the terminal designation of your Aquatos data logger. Refer to the corresponding logger manual for the correct terminal assignment.

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Note: Multiple sensors can be operated in parallel on the SDI-12 bus (the Aquatos data logger supports up to 24 measurement channels) (max. total cable length 500 m). Each sensor requires a unique SDI-12 address (0–9, A–Z, a–z). The factory address of the Type 0420 is ‘0’.

3.4 Initial Start-Up

5. Connect the sensor to the data logger as described in Section 3.3.
6. Verify correct seating and cable routing.
7. Supply the data logger with power. The sensor requires no separate power supply — it is powered via the data logger’s SDI-12 bus.
8. Configure the sensor using the Bluetooth app (see Part 4).
9. Start a test measurement and verify the plausibility of the readings.

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Part 4 Configuration

The Ceramic Pressure Sensor Type 0420 is configured using the Bluetooth app. Bluetooth app is a browser-based application that communicates directly with the sensor via Bluetooth 5 (BLE). No app installation is required.

4.1 Opening the Bluetooth app

Open the Bluetooth app in a supported browser:

Platform	Supported Browser
Windows / macOS / Linux	Google Chrome, Microsoft Edge
Android	Google Chrome, Microsoft Edge
iOS (iPhone / iPad)	Bluefy Browser (free on App Store)

URL: https://ltxdata.net/ttblx/

Note: Firefox and Safari do not support Web Bluetooth and are therefore not suitable.

4.2 Establishing a Connection

10. Ensure the sensor is powered (data logger switched on or test supply connected).
11. Enable Bluetooth on your device.
12. Open https://ltxdata.net/ttblx/ in a supported browser.
13. Click “Connect device”. A list of available devices appears.
14. Select the Ceramic Pressure Sensor Type 0420 from the list.
15. When prompted, enter the device-specific BLE PIN (printed on the sensor or in the delivery note). Default: 0000.

Caution: The PIN is never transmitted in plain text. After multiple incorrect entries, the device temporarily blocks the BLE connection.

4.3 Setting Measurement Interval and Parameters

In the “Settings” section, you can adjust the following parameters:

- Measurement interval (typically 1, 5, 15 or 60 minutes)
- Depth offset: distance from sensor to reference level (for direct gauge reading in m a.s.l.)
- Temperature compensation: enable or disable
- Device name (for easier identification with multiple sensors)

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4.4 Configuring the SDI-12 Address

If multiple sensors are operated on one data logger, each sensor must have a unique SDI-12 address. The address is configured in the Bluetooth app under “Terminal”:

- Send the SDI-12 command “0AC!” to change the address from ‘0’ to ‘C’ (example).
- Valid addresses: 0–9, A–Z, a–z

Note: The SDI-12 address can also be changed via the data logger, provided it supports SDI-12 address change commands. Refer to the data logger manual.

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Part 5 Operation

5.1 Starting Measurements

After completing configuration, the sensor automatically starts measuring at the set interval. No further steps are required.

5.2 Reading Measurement Values

On SDI-12 polling, the Type 0420 transmits the following measured values:

Measured value	Unit / Description
Pressure (relative)	bar (reference: atmospheric pressure via capillary)
Water depth (calculated)	m (from pressure, configurable)
Temperature	°C (water temperature)

The measured values are polled by the connected data logger at the set intervals, stored and — depending on the logger type — transmitted via mobile network to the Sensormanager web portal.

5.3 Operation with Aquatos Data Loggers

The Ceramic Pressure Sensor Type 0420 is approved for operation with the following Aquatos data loggers from TerraTransfer:

Data Logger	Description
Aquatos Web LTX	Online logger with LTE-M mobile network. Data transmission and remote access via Sensormanager web portal.
Aquatos mini	Offline BLE logger for local readout via smartphone or tablet.

> **Note:** The Aquatos nano is not recommended as it has a permanently integrated pressure sensor and does not support an external pressure probe.

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Part 6 Maintenance

6.1 Cleaning the Ceramic Membrane

Depending on water quality and service duration, deposits, biofilm or limescale may form on the membrane. Clean the sensor regularly (recommended: once a year; more frequently in heavily loaded water).

16. Remove the sensor from the measurement tube.
17. Scrub deposits with clean water and a soft brush (e.g. toothbrush). The ceramic membrane is scratch-resistant and tolerates a soft brush without any problems.
18. For limescale deposits: briefly soak in diluted acetic acid (max. 5%), then rinse thoroughly with tap water.
19. Allow the sensor to dry and check for visible damage.

Caution: Do not use abrasives, solvents or high-pressure cleaners. Do not use hard or sharp tools near the membrane. Despite the scratch-resistance of the ceramic, seals and cable routing can be damaged by rough cleaning.

6.2 Cable and Capillary

Inspect the connection cable annually for kinks, pressure marks and damage to the cable seal. The capillary inside the cable must always remain free and air-permeable.

- Cable outlet at the sensor: check for tightness and secure fit
- Capillary end at logger connection: keep dry and open
- No water at capillary inlet (e.g. from condensation in the control cabinet)

6.3 Calibration Interval

The sensor leaves the factory calibrated. For metrologically critical applications, we recommend a field calibration against a known water level every 2–3 years. Contact TerraTransfer for calibration enquiries.

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Part 7 Troubleshooting

Symptom	Possible Cause / Action
No SDI-12 signal	Check cable connection. Measure supply voltage (5–16 V). Check SDI-12 address in Bluetooth app.
Constant or implausible reading	Capillary pinched or blocked. Check cable for kinks; replace cable if necessary.
Reading fluctuates irregularly	Water has entered capillary. Remove sensor, check capillary end.
BLE connection not possible	Distance to sensor exceeds 10 m (range). Is browser Bluetooth enabled? Use supported browser.
PIN forgotten / blocked	Leave sensor without supply voltage for 10 minutes, then retry PIN entry. For permanent lock: return to TerraTransfer.
Temperature reading incorrect	Check thermistor offset (Bluetooth app » Settings » Temperature offset). If deviation >1 °C: calibration recommended.

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Part 8 Repair and Return

Repairs to the Ceramic Pressure Sensor Type 0420 may only be carried out by TerraTransfer GmbH or authorised specialist personnel. Unauthorised modifications to the device will void the warranty.

Return Procedure

20. Contact TerraTransfer by phone or e-mail before returning the device to obtain an RMA number.
21. Package the device securely with adequate cushioning. Ensure the sensor is dry and clean.
22. Enclose a brief fault description and your contact details.
23. Send the device to the TerraTransfer service address (see Contact).

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Part 9 Technical Data

9.1 Mechanical

Property	Value
Housing material	Stainless steel
Pressure membrane	Ceramic
Cable length	Individual per order (permanently fitted)
Protection class	IP68
Minimum tube diameter	¾-inch (19.05 mm)

9.2 Electrical / Communication

Property	Value
Supply voltage	2.8–16 V DC (SDI-12 bus)
Quiescent current	<10 µA
Interface	SDI-12 V1.3
Short-range wireless	Bluetooth 5 (BLE), TerraTransfer sensor platform
BLE security	PIN protection, TLS
BLE range	approx. 10 m in open field (environment-dependent)
Configuration	Bluetooth app (browser-based, no app download)
Factory SDI-12 address	‘0’

9.3 Measurement

Property	Value
Pressure type	Relative pressure (with venting capillary)
Pressure measurement range	0–1 bar (0–10 mWS)
Accuracy	±0.5 % FS (typical)
Temperature measurement	Yes, integrated
Temperature range	−20 to +80 °C
Temperature accuracy	±0.3 °C
Long-term stability	Excellent (ceramic membrane corrosion-resistant)

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9.4 Environmental

Property	Value
Operating temperature	−20 to +80 °C
Storage temperature	−40 to +80 °C
Protection class	IP68 (continuous submersion)
Medium	Water (including mineral-rich, mildly aggressive)

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Part 10 Operator Obligations and Disposal

Operator Obligations

The operator is obliged to use the sensor in accordance with this manual and to operate it only in perfect condition. Defective devices must be taken out of service immediately.

Disposal (EU WEEE Directive)

In accordance with the EU WEEE Directive (2012/19/EU), waste electrical equipment must be disposed of separately from household waste. Return old devices to municipal collection points or to TerraTransfer GmbH for proper disposal.

> **Caution:** The device must not be disposed of via normal household waste. Improper disposal may cause environmental damage.

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Contact

For questions about this product, service or spare parts, please contact:

TerraTransfer GmbH

Ottostraße 19a

D-44867 Bochum, Germany

Phone: +49 (0)2327 83 44 85-1

Fax: +49 (0)2327 83 44 85-7

E-mail: info@terratransfer.de

Web: www.terratransfer.de

Commercial register: Amtsgericht Bochum HRB 6778

Managing Director: Marcel Delker

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